ECE 1212 – Electronic Circuit Design Laboratory
Experiment #2 – Operational Amplifier Applications

Introduction

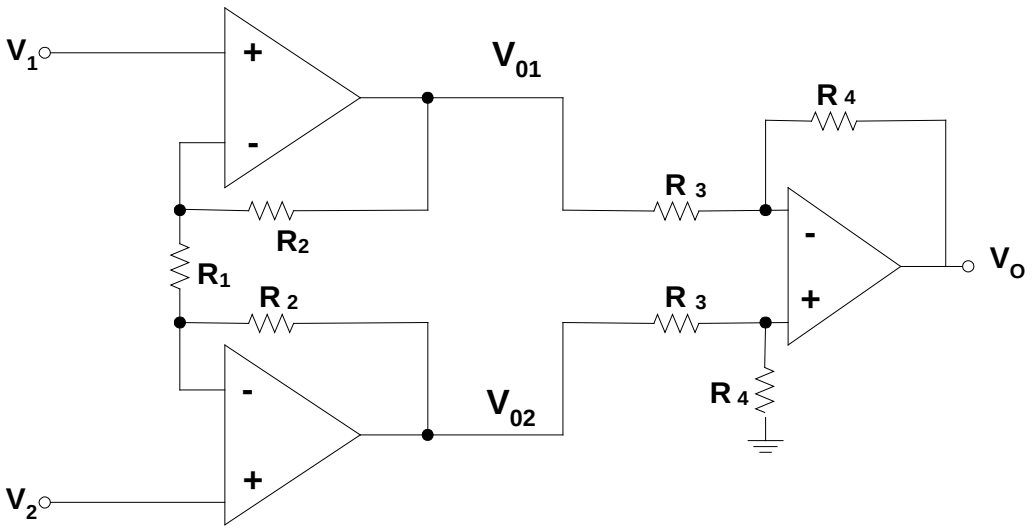
In this experiment you will design, build and test several circuits, which use op-amps as central element. Your plans should include designs, test procedures and expected results for each circuit. Remember to consider any real (as opposed to ideal) properties of the op-amp, which may affect your designs. The lab will be split into 3 parts: Instrumentation Amplifier, II. Ramp Generation, III. Simulation.

Deliverables

1. Prelab 2A (part I – 2 prompts, part II – 1 prompt)
2. Prelab 2B (part III – 1 prompt)
3. Lab Report

Part I. Instrumentation Amplifier

We can augment the design of the difference amplifier to achieve a high input impedance and to make it easier to set the gain. Consider the following circuit, where you should use ±12V power supplies for all op amps:



In the Pre-Lab (2A):

(a) Analyze this circuit and show that the output voltage can be expressed in the following form:

$$V_O = K\,(V_2 - V_1)$$

or

$$V_0 = -\frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1)$$

(b) Design the circuit for K = -3. Use $R_2 = R_3 = R_4 = 1K\Omega$. What value of R_1 should you choose?

In the Laboratory:

(a) Ground V_1 and connect a function generator to V_2. Set the input signal as a sinusoidal waveform with frequency of 2 kHz and amplitude of 1 V (peak to peak). Observe the output. Copy the scope picture and specify the input and output peak-to-peak values. Finally, calculate the amplification factor.

(b) Measure the input impedance seen at V_2 with V_1 grounded. What value do you expect to measure? Recall from Experiment 1 the difficulties that are associated with measuring R_{in} when using an AC source at the input. Use a DC voltage at V_2 for this measurement, so no frequency dependent.

Note 1: The nominal input resistance of the multimeter (when used as a DC voltmeter) is 10 MΩ, and the input resistance of a 741 op-amp is also in the MΩ range. Also, the input resistance of the instrumentation amplifier will be well into the GΩ range, for the reasons explained in the lecture notes for Experiment 1.

Therefore, if you use the multimeter to measure the voltage across your instrumentation amplifier, these two devices will be in parallel, and this can significantly affect your input impedance measurement. One solution to this problem is to reprogram the multimeter to have a much higher input resistance (>10 GΩ). Ask the instructor or TA if you need help with this.

Note 2: The input impedance has been observed to be dependent on the applied DC voltage. If you experience this phenomenon, make several measurements for different DC voltages, and report your results.

(c) Ground V_1 and connect a function generator to V_2. Set the input signal as a sinusoid waveform with frequency of 2 kHz and amplitude 1 V (peak-to-peak). Observe the output. Copy the scope picture, specify the input and output peak-to-peak values, and calculate the amplification factor. Finally, replace R_1 with a fixed resistor in series with a potentiometer to set K to exactly -3.0.

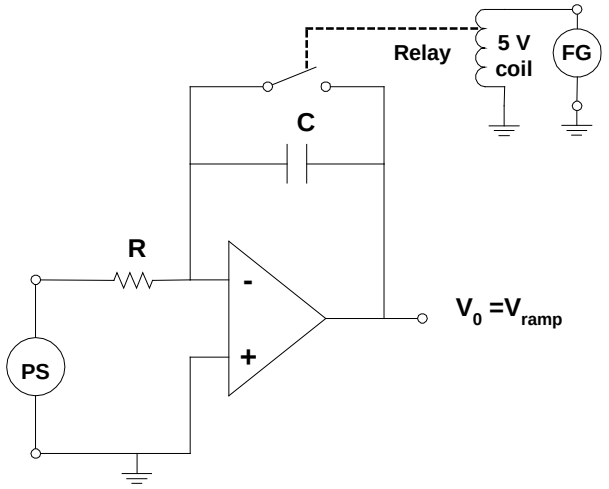
(d) Connect the two outputs of the function generator as V_1, V_2. The following chart shows the input parameters. Observe the output from scope. Show your scope pictures, and specify peak-to-peak voltage values.

	Inputs 1	Inputs 2
V_1	Sine, 1KHz, 1Vpp	Sine, 1 KHz, 1Vpp
V_2	Sine, 1KHz, 2Vpp	Sine, 1 KHz, 1Vpp

(e) For Inputs 2 in step 5 ($V_1 = V_2$ = 1 Vpp, 1 KHz), observe the effects of a phase difference between the two inputs on the output of your instrumentation amplifier. (When the two outputs of the signal generator are synchronized, you can set the phase difference between them.) How does the amplifier output voltage help you to predict phase difference between the two input signals?

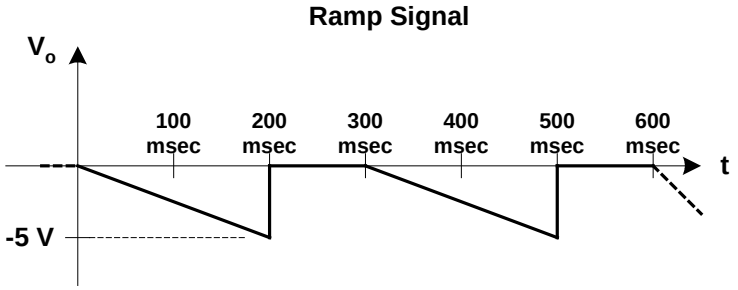
Part II. Ramp Generation

We will look at a method we can use to generate a special form of signal, termed a *ramp* (Ref. Sedra and Smith: 4e. 72-78, 5e. 107-112, or 6e. 82-86). A circuit to do this is given below. Use ±12Vdc for the op-amp supply voltages.



PS is a constant voltage source.

The circuit uses an electromechanical switching device called a relay. There are fully electronic ways to replace the relay. Note that without the relay, the circuit would integrate a dc input signal indefinitely until the 12 volt supply voltage is exceeded. Even if the input voltage goes to zero there would be no way to reduce the charge stored on the capacitor! Your goal in this part of the experiment is to realize the above ramp signal at the output of the circuit:



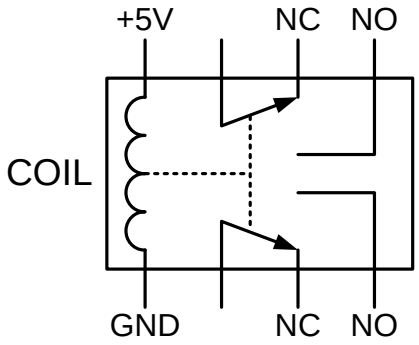
In the Pre-Lab (2A):

Analyze the circuit using basic KVL real time analysis for the case when the initial charge on the capacitor is zero and the relay contact is "open". This is as if the capacitor has been shorted (by the relay contact) for a "long" period of time and the relay contacts are opened at time 0. This analysis should help you to see how to pick R and C values, how to set up the FG, and how to select the dc voltage setting of the PS in order to obtain -5 Vdc in 0.2 seconds.

In the Laboratory:

Build your design and then fine-tune your design choices to meet the specification as well as you can. Relays are fairly slow, taking a few milliseconds to open and close. How do these time delays affect your output signal? Record the scope trace showing the ramp (make sure your scope is under DC coupling not AC coupling, you can go to channel setting to change it to DC coupling if it is under AC coupling).

Note: The following figure shows the pin diagram for the relay in your parts kit. The function generator must be connected across the coil terminals, and must be set to provide a waveform appropriate for switching the relay on and off. (What wave shape should you use? What should the frequency, amplitude and DC offset be? What should the duty cycle be, and why?) You should connect the pole (slider) and either the NC and NO terminal from one side of the relay across the capacitor. (How does your choice of either the NC or NO terminal affect your choice of duty cycle?) Also, the connection by which the coil causes the relay to open and close (indicated by a dashed line in the circuit diagram for the ramp generator) is made inside the relay, so there is no need for you to make this connection with an external wire.



Part III – Simulation

In the pre-lab 2B:

Build your instrumentation amplifier as shown Part I Lab part (d).

In the Lab:

This part is a simulation part. Design your circuit and simulate your circuit in PSPICE/Multisim for part I Instrumentation Amplifier part I lab (d). To compare your measured result and the simulated result and explain the difference.

PSpice/Multisim Instructions

Use LM741/NS op-amps from the NAT_SEMI library.
These devices have the following terminal connections:

- Pin 1: Offset Null
- Pin 2: Inverting input
- Pin 3: Non-inverting input
- Pin 4: Negative power supply
- Pin 5: Offset Null
- Pin 6: Output
- Pin 7: Positive Power Supply

